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(54) **REVOKED FIRMWARE ROLLBACK PREVENTION**

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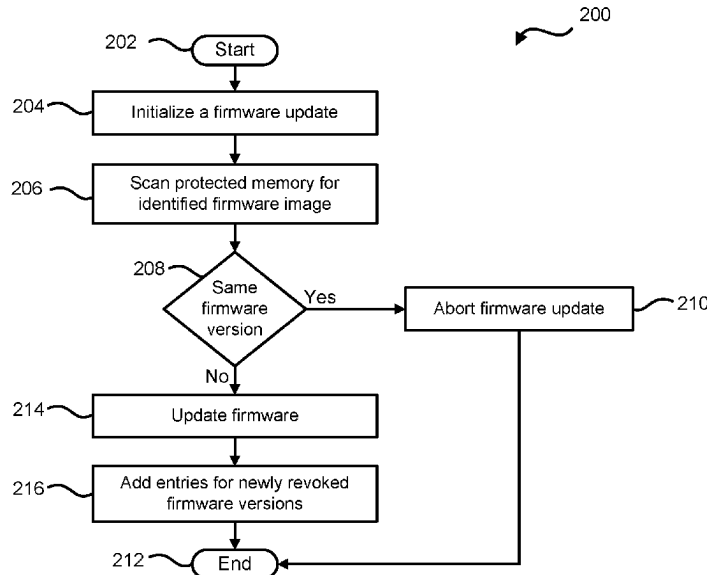
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(57) **ABSTRACT**

An information handling system includes multiple components including a first component. The first component includes a protected memory and a basic input/output system (BIOS). The protected memory stores a revoked versions list. The BIOS initializes a firmware update for a firmware image having a firmware version. The BIOS scans the revoked versions list for the firmware version of the firmware image. In response to the firmware version not being located within the revoked versions list, the BIOS completes the firmware update, and determines whether a revoked firmware version is included in the firmware update. In response to the revoked firmware version being included in the firmware update, the BIOS adds an entry in the revoked versions list. The entry is associated with the revoked firmware version included in the firmware update.

20 Claims, 4 Drawing Sheets



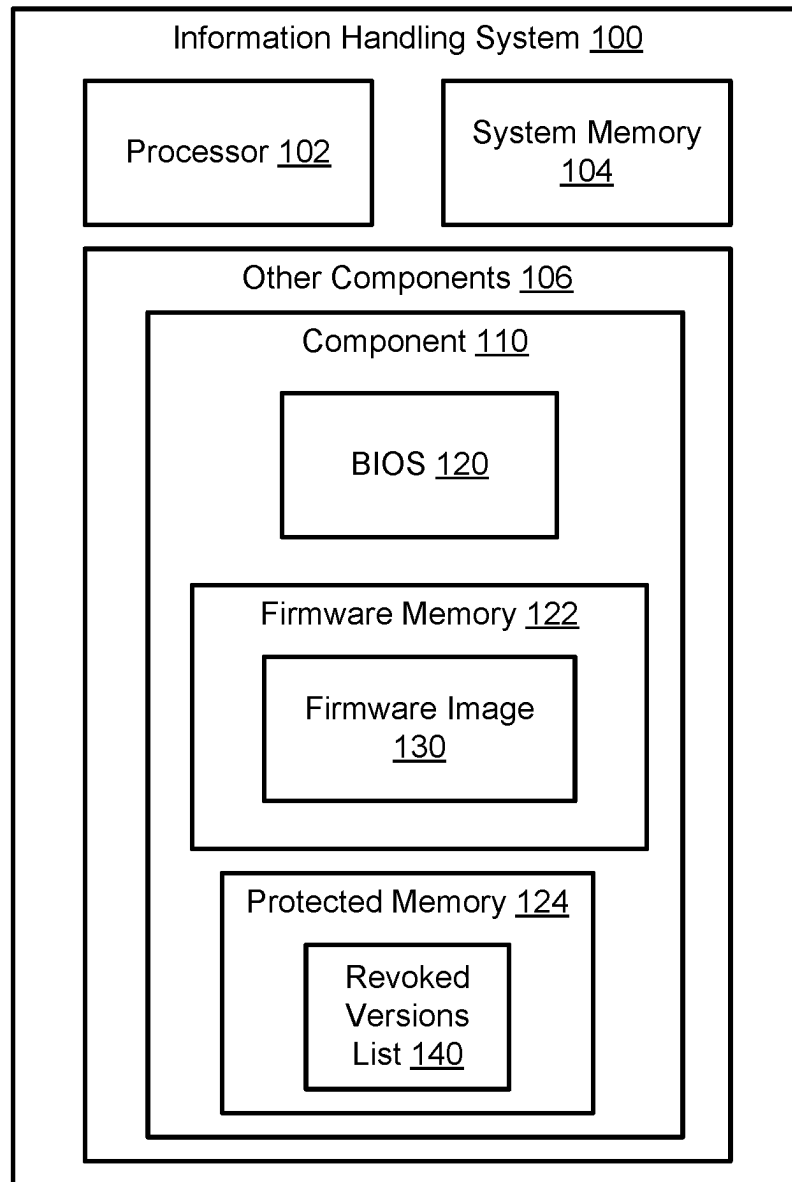
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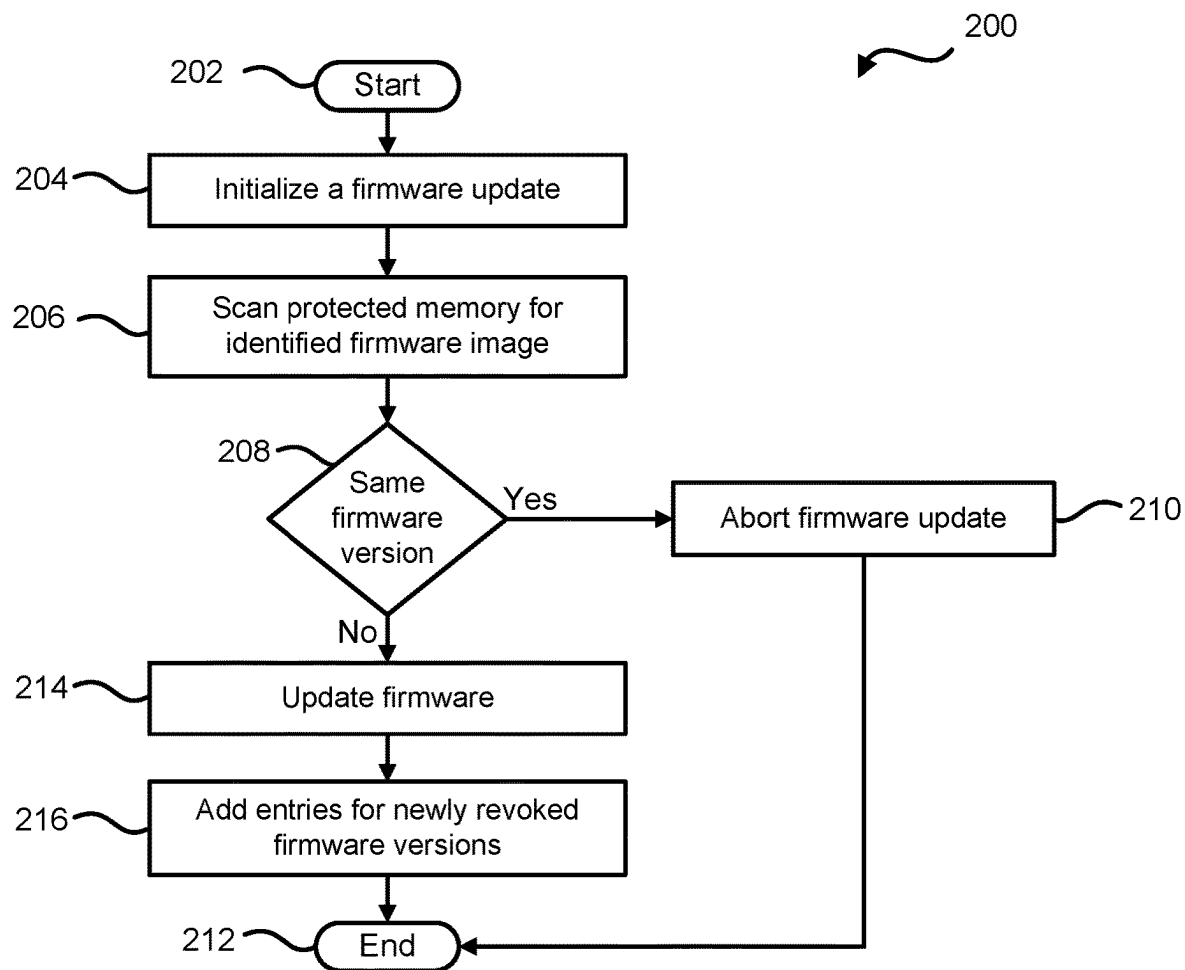
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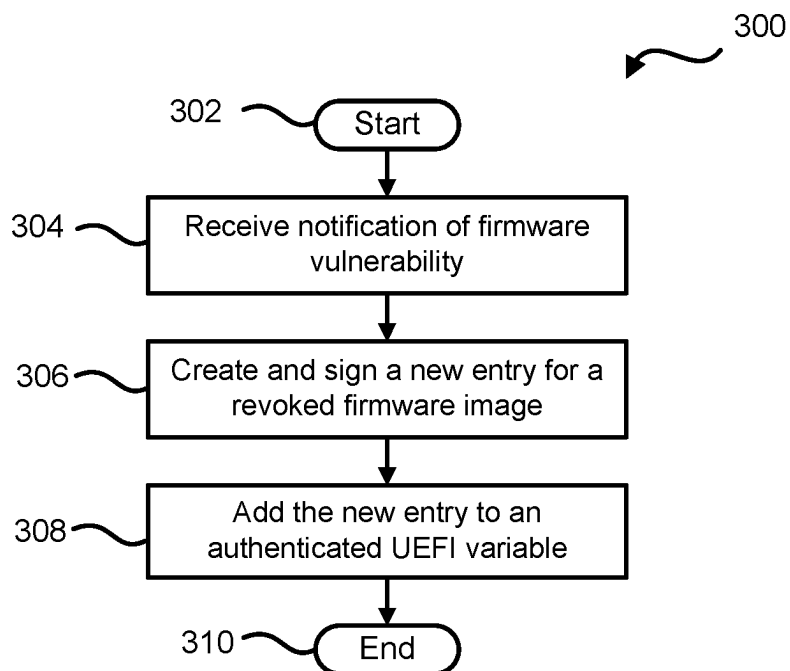
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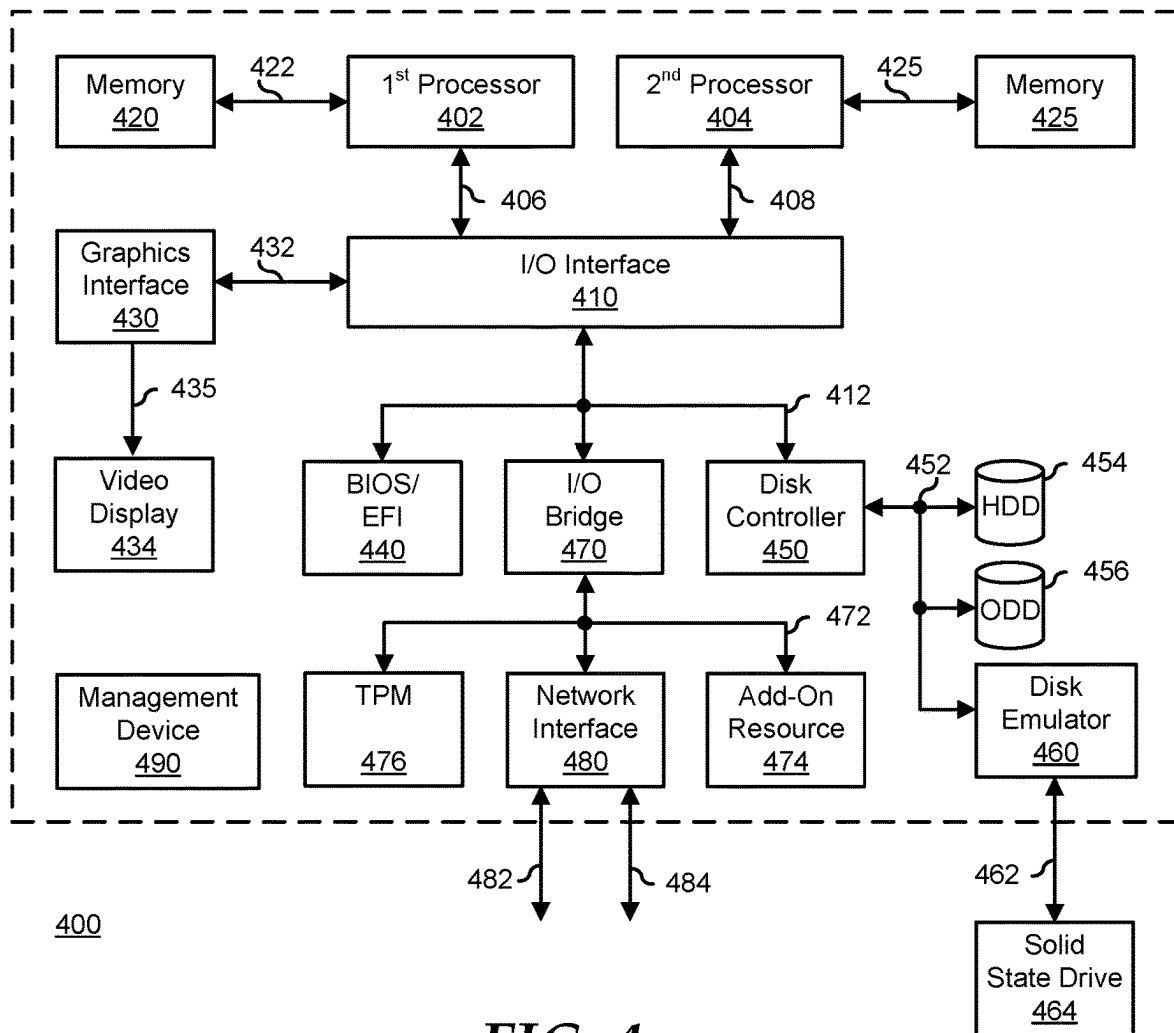
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**FIG. 1**

**FIG. 2**

**FIG. 3**

**FIG. 4**

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REVOKED FIRMWARE ROLLBACK PREVENTION

FIELD OF THE DISCLOSURE

The present disclosure generally relates to information handling systems, and more particularly relates to revoked firmware rollback prevention.

BACKGROUND

As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs, and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

An information handling system has a first component that includes a protected memory and a basic input/output system (BIOS). The protected memory stores a revoked versions list. The BIOS initializes a firmware update for a firmware image having a firmware version. The BIOS scans the revoked versions list for the firmware version of the firmware image. In response to the firmware version not being located within the revoked versions list, the BIOS completes the firmware update, and determines whether a revoked firmware version is included in the firmware update. In response to the revoked firmware version being included in the firmware update, the BIOS adds an entry in the revoked versions list. The entry is associated with the revoked firmware version included in the firmware update.

BRIEF DESCRIPTION OF THE DRAWINGS

It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

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FIG. 1 is a diagram of a portion of an information handling system according to at least one embodiment of the present disclosure;

FIG. 2 is a flow diagram of a method for preventing a firmware rollback to revoked firmware versions according to at least one embodiment of the present disclosure;

FIG. 3 is a flow diagram of another method for preventing a firmware rollback to revoked firmware versions according to at least one embodiment of the present disclosure; and

FIG. 4 is a block diagram of a general information handling system according to an embodiment of the present disclosure.

The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

FIG. 1 illustrates a portion of an information handling system **100** according to an embodiment of the present disclosure. For purposes of this disclosure, an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (such as a desktop or laptop), tablet computer, mobile device (such as a personal digital assistant (PDA) or smart phone), server (such as a blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

Information handling system **100** includes a processor **102**, system memory **104**, and other components **106**. Other components **106** may include any number of components, but a single component **110** has been illustrated for clarity and brevity. Component **110** includes a basic input/output system (BIOS) **120**, a firmware memory **122**, and a protected memory. Firmware memory **122** can include a non-volatile random access memory (NVRAM), and is configured to store a firmware image **130**. Firmware image **130** generally includes a collection of firmware routines, device drivers, and other software programs. Protected memory **124** may store a revoked versions list **140**. Protected memory **124** may be any suitable memory that secures the stored data from unauthorized access. In this situation, unauthorized software/firmware is prevented from accessing the data or

content within protected memory **124**. In an example, the revoked versions list **140** may be any suitable secured data such as a unified extensible firmware interface (UEFI) authenticated variable. In an example, the other components **106** may include, but are not limited to, network interface cards, memory drives, memory controllers, and redundant array of independent disks (RAID) controllers. Information handling system **100** may include additional components without varying from the scope of this disclosure.

In certain examples, while FIG. **1** illustrates only a single component **110** with a single firmware memory **122**, a single firmware image **130**, a single BIOS **120**, a single protected memory **124**, and a single revoked versions list **140**, each of the other components **106** may include a firmware memory, a firmware image **130**, a BIOS, a protected memory, and a revoked versions list. In this situation, firmware updates for a particular component may be provided to that component so that the BIOS of the component may perform the firmware update on the component. For clarity and brevity, firmware updates and revoked versions list **140** will be described only with respect to firmware image **130** and revoked versions list **140**, but the same operations may be performed in each of the other components **106** without varying from the scope of this disclosure.

During operation of information handling system **100**, one or more firmware updates may be performed by BIOS **120**. In certain examples, firmware updates for each of the other components **106** may be managed by different entities, and each of these different entities may also be different than the entity to manage firmware updates for main devices within information handling system **100**. In an example, a security vulnerability or security flaw may be determined for firmware image **130**, such that the firmware image may need to be updated to correct the security vulnerability.

In previous information handling systems, if a security vulnerability is found in a firmware of a component a firmware vendor or entity may provide a new firmware version to correct or remove the vulnerability. In response to the new or updated firmware version being received, the previous information handling system may perform the firmware update to install the new version of the firmware. However, the previous information handling systems may have another potential attack vector, such as an attacker may be able to downgrade the firmware to the version with the known vulnerability and then exploit the vulnerability. Information handling system **100** may be improved by processor **102** and BIOS **120** preventing a rollback or downgrading of a firmware version from a currently installed version to a version having a known vulnerability as will be described herein.

During a firmware update, BIOS **120** may scan revoked versions list **140** within protected memory **124** for the firmware version of the initialized firmware update. In an example, a signed update utility within BIOS **120** may perform the scan of revoked versions list **140**. In an example, the firmware update package may include an identity of revoked firmware versions. In certain examples, a revoked firmware version may be identified by a firmware version number or the like. In an example, the revoked firmware version may be any previous firmware version, such as a currently installed firmware version being replaced by the firmware update or any other previous firmware version no matter how many updates have been made since that version. In an example, revoked versions list **140** may contain one or more revoked version numbers including three or four previous version numbers. In this example, the firm-

ware versions in revoked version list **140** are never removed even when new revoked firmware versions are added to the revoked version list.

Based on the scan of revoked versions list **140**, BIOS **120** may determine whether the firmware version for the initialized firmware update is located within the revoked versions list of protected memory **124**. If the firmware version for the initialized firmware update is located within the protected memory, BIOS **120** may abort the firmware update. If the firmware version for the initialized firmware update is not located within the protected memory, BIOS **120** may perform one or more operations to complete the firmware update. For example, one of the operations may include BIOS **120** creating one or more entries for any newly revoked firmware versions. In this example, the firmware update may include the one or more revoked firmware version numbers. After the new entries have been created, BIOS **120** may store the new entries in revoked versions list **140**. These new entries and all other entries in revoked versions list **140** may be utilized by BIOS **120** during any future firmware updates.

In certain examples, a notification of a firmware vulnerability may be provided to information handling system **100**, to a user of the information handling system, or the like. In an example, the notification may be received by information handling system **100** in any suitable manner, such as a push notification from a firmware vulnerability tracking component, an input from a user of the information handling system, or the like. The firmware vulnerability notification may include any suitable data such as the firmware version number associated with the vulnerability, or the like. Based on the data in the notification, processor **102** or a processor of component **110** may create a revoked firmware entry associated with the firmware vulnerability. Processor **102** may provide the revoked firmware entry to component **110**, which in turn may add the entry to revoked versions list **140**.

In an example, the new entry is added to an authenticated UEFI variable of revoked versions list **140** in any suitable manner. In particular, the authenticated UEFI variable, such as revoked versions list **140**, may be protected by private-public key pairs. In this situation, a private key for the authenticated UEFI variable may be stored in any suitable memory. For example, the private key may be stored within a secure memory in a trust platform manager (TPM) of information handling system **100**, within a key management system (KMS), within a server including a hardware security module (HSM), or the like. In an example, the private key may be utilized to modify, add, or remove revoked firmware versions from the UEFI variable, such as revoked versions list **140**. Based on the private key for the authenticated UEFI variable, a user may utilize the private key to revoke firmware versions that have a vulnerability. In this example, the user may immediately revoke firmware versions identified with vulnerability without waiting for an entity associated with a component of the firmware to send a firmware update.

FIG. **2** shows a method **200** for preventing a firmware rollback to revoked firmware versions according to at least one embodiment of the present disclosure, starting at block **202**. In an example, the method **200** may be performed by any suitable component including, but not limited to, a processor, such as processor **102** of FIG. **1**. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary, and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure.

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At block **204**, a firmware update is initialized. In an example, the firmware update may be for firmware of any suitable component within an information handling system including, but not limited to, a network interface card, a memory drive, a memory controller, and a redundant array of independent disks (RAID) controller. In certain examples, the firmware update for the component may be managed by a different entity as compared to the entity to manage motherboard processor firmware updates.

At block **206**, a protected memory is scanned for the firmware version of the initialized firmware update. In certain examples, the protected memory may be any suitable type of memory, such as a non-volatile random access memory (NVRAM) or the like, which prevents unauthorized modification of the memory contents. In an example, a list of revoked firmware versions may be stored in the protected memory of the information handling. The list of revoked firmware versions may be stored as a universal extensible firmware interface (UEFI) authenticated variable in the protected memory.

At block **208**, a determination is made whether a firmware version for the initialized firmware update is located within the protected memory, such as in the list of revoked firmware versions. If the firmware version for the initialized firmware update is located within the protected memory, the firmware update is aborted at block **210** and the flow ends at block **212**.

If the firmware version for the initialized firmware update is not located within the protected memory, the firmware update is completed at block **214**. In an example, the firmware update package may include an identity of revoked firmware versions. In certain examples, a revoked firmware version may be identified by a firmware version number or the like. In an example, the revoked firmware version may be any previous firmware version, such as a currently installed firmware version being replaced by the firmware update or any other previous firmware version no matter how many updates have been made since that version. At block **216**, entries for any newly revoked firmware versions are added to the protected memory, such as in the list of revoked firmware versions, and the flow ends at block **212**.

FIG. 3 shows a method **300** for preventing a firmware rollback to revoked firmware versions according to at least one embodiment of the present disclosure, starting at block **302**. In an example, the method **300** may be performed by any suitable component including, but not limited to, a processor, such as processor **102** of FIG. 1. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary, and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure.

At block **304**, a notification of a firmware vulnerability is received. In certain examples, the notification may be received in any suitable manner, such as a push notification from a firmware vulnerability tracking component, an input from a user of the information handling system, or the like. At block **306**, a new entry for a revoked firmware image version is created and signed within the information handling system. At block **308**, the new entry is added to an authenticated UEFI variable, and the flow ends at block **310**.

In an example, the new entry is added to the authenticated UEFI variable in any suitable manner or set of operations. In particular, the authenticated UEFI variable may be protected by private-public key pairs. In this situation, a user of the information handling system needs to hold the private key for the authenticated UEFI variable before the user may

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modify, add, or remove revoked firmware versions from the UEFI variable list. Based on the user holding the private key for the authenticated UEFI variable, the user may utilize the private key to sign new entries for the authenticated UEFI variable. The user may then add such signed entries to the variable, thus revoking firmware versions that have a vulnerability. In this example, the user may immediately revoke firmware versions identified with vulnerability without waiting for an entity associated with a component of the firmware to send a firmware update.

FIG. 4 shows a generalized embodiment of an information handling system **400** according to an embodiment of the present disclosure. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **400** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **400** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **400** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **400** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **400** can also include one or more buses operable to transmit information between the various hardware components.

Information handling system **400** can include devices or modules that embody one or more of the devices or modules described below and operates to perform one or more of the methods described below. Information handling system **400** includes a processors **402** and **404**, an input/output (I/O) interface **410**, memories **420** and **425**, a graphics interface **430**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **440**, a disk controller **450**, a hard disk drive (HDD) **454**, an optical disk drive (ODD) **456**, a disk emulator **460** connected to an external solid state drive (SSD) **462**, an I/O bridge **470**, one or more add-on resources **474**, a trusted platform module (TPM) **476**, a network interface **480**, a management device **490**, and a power supply **495**. Processors **402** and **404**, I/O interface **410**, memory **420**, graphics interface **430**, BIOS/UEFI module **440**, disk controller **450**, HDD **454**, ODD **456**, disk emulator **460**, SSD **462**, I/O bridge **470**, add-on resources **474**, TPM **476**, and network interface **480** operate together to provide a host environment of information handling system **400** that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system **400**.

In the host environment, processor **402** is connected to I/O interface **410** via processor interface **406**, and processor **404** is connected to the I/O interface via processor interface **408**. Memory **420** is connected to processor **402** via a memory interface **422**. Memory **425** is connected to processor **404** via a memory interface **427**. Graphics interface **430** is connected to I/O interface **410** via a graphics interface **432** and provides a video display output **436** to a video display **434**. In a particular embodiment, information handling system **400** includes separate memories that are dedicated to each of processors **402** and **404** via separate memory interfaces. An example of memories **420** and **430** include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof.

BIOS/UEFI module **440**, disk controller **450**, and I/O bridge **470** are connected to I/O interface **410** via an I/O channel **412**. An example of I/O channel **412** includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface **410** can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I²C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module **440** includes BIOS/UEFI code operable to detect resources within information handling system **400**, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module **440** includes code that operates to detect resources within information handling system **400**, to provide drivers for the resources, to initialize the resources, and to access the resources.

Disk controller **450** includes a disk interface **452** that connects the disk controller to HDD **454**, to ODD **456**, and to disk emulator **460**. An example of disk interface **452** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **460** permits SSD **464** to be connected to information handling system **400** via an external interface **462**. An example of external interface **462** includes a USB interface, an IEEE 4394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive **464** can be disposed within information handling system **400**.

I/O bridge **470** includes a peripheral interface **472** that connects the I/O bridge to add-on resource **474**, to TPM **476**, and to network interface **480**. Peripheral interface **472** can be the same type of interface as I/O channel **412** or can be a different type of interface. As such, I/O bridge **470** extends the capacity of I/O channel **412** when peripheral interface **472** and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel **472** when they are of a different type. Add-on resource **474** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **474** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **400**, a device that is external to the information handling system, or a combination thereof.

Network interface **480** represents a NIC disposed within information handling system **400**, on a main circuit board of the information handling system, integrated onto another component such as I/O interface **410**, in another suitable location, or a combination thereof. Network interface device **480** includes network channels **482** and **484** that provide interfaces to devices that are external to information handling system **400**. In a particular embodiment, network channels **482** and **484** are of a different type than peripheral channel **472** and network interface **480** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **482** and **484** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **482** and **484** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

Management device **490** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device (CPLD), and the like, which operate together to provide the management environment for information handling system **400**. In particular, management device **490** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I²C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **400**, such as system cooling fans and power supplies. Management device **490** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **400**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **400**.

Management device **490** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **400** when the information handling system is otherwise shut down. An example of management device **490** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like. Management device **490** may further include associated memory devices, logic devices, security devices, or the like, as needed or desired.

Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments

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of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover any and all such modifications, enhancements, and other embodiments that fall within the scope of the present invention. Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents and shall not be restricted or limited by the foregoing detailed description.

What is claimed is:

1. An information handling system comprising:
a protected memory to store a revoked versions list;
a processor configured to modify a revoked firmware version of a first firmware image in the revoked versions list; and
a basic input/output system (BIOS), the BIOS configured to:
initialize a firmware update for a firmware image having a firmware version;
scan the revoked versions list for the firmware version of the firmware image; and
in response to the firmware version not being located within the revoked versions list, the BIOS further configured to:
complete the firmware update;
determine whether a firmware version number for a new revoked firmware version is included in the firmware update, wherein the new revoked firmware version is a previous firmware version, wherein multiple firmware updates occurred between the previous firmware version and the firmware update; and
in response to the firmware version number for the new revoked firmware version being included in the firmware update, add an entry in the revoked versions list, the entry being associated with the firmware version number for the new revoked firmware version included in the firmware update.
2. The information handling system of claim 1, wherein the processor further configured to:
receive a notification of a firmware version vulnerability associated with a first firmware version;
create a new entry for the first firmware version; and
add the new entry to the revoked versions list.
3. The information handling system of claim 2, wherein the new entry is signed by a private key prior to being added to the revoked versions list.
4. The information handling system of claim 1, wherein the scan of the revoked versions list is performed by a signed update utility.
5. The information handling system of claim 1, wherein the new revoked firmware version in the firmware update is for versions prior to a current firmware image.
6. The information handling system of claim 1, wherein the new revoked firmware version has a security vulnerability.
7. The information handling system of claim 1, wherein the new revoked firmware version is stored in a unified extensible firmware interface variable.

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8. The information handling system of claim 1, in response to the firmware version being located within the revoked versions list, the BIOS further configured to abort the firmware update.

9. A method comprising:
modifying, by a processor, a revoked firmware version of a first firmware image in a revoked versions list;
initializing, by a basic input/output system (BIOS), a firmware update for a firmware image having a firmware version;
scanning, by the BIOS, the revoked versions list for the firmware version of the firmware image; and
in response to the firmware version not being located within the revoked versions list:
completing the firmware update;
determining whether a firmware version number for a new revoked firmware version is included in the firmware update, wherein the new revoked firmware version is a previous firmware version, wherein multiple firmware updates occurred between the previous firmware version and the firmware update; and
in response to the firmware version number for the new revoked firmware version being included in the firmware update, adding an entry in the revoked versions list, the entry is associated with the firmware version number for the new revoked firmware version included in the firmware update.
10. The method of claim 9, further comprising:
receiving, by the processor, a notification of a firmware version vulnerability associated with a first firmware version;
creating, by the processor, a new entry for the first firmware version; and
adding the new entry to the revoked versions list.

11. The method of claim 10, wherein the new entry is signed by a private key prior to being added to the revoked versions list.

12. The method of claim 9, wherein the scanning of the revoked versions list is performed by a signed update utility within the BIOS.

13. The method of claim 9, wherein the new revoked firmware version in the firmware update is for a current firmware image.

14. The method of claim 9, wherein the new revoked firmware version has a security vulnerability.

15. The method of claim 9, wherein the new revoked firmware version is a unified extensible firmware interface variable.

16. The method of claim 9, wherein in response to the firmware version being located within the revoked versions list:

aborting the firmware update.

17. A method comprising:
modifying, by a processor, a revoked firmware version of a first firmware image in a revoked versions list;
initializing, by a basic input/output system (BIOS), a firmware update for a firmware image having a firmware version;
scanning, by the BIOS, the revoked versions list for the firmware version of the firmware image;
if the firmware version is not within the revoked versions list, then:
completing the firmware update;
determining whether a firmware version number for a new revoked firmware version is included in the firmware update, wherein the new revoked firmware

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version is a previous firmware version, wherein multiple firmware updates occurred between the previous firmware version and the firmware update; and

if the firmware version number for the new revoked 5
firmware version is in the firmware update, then adding an entry in the revoked versions list, the entry is associated with the firmware version number for the new revoked firmware version included in the firmware update, wherein the new revoked firmware 10
version has a security vulnerability; and

if the firmware version is within the revoked versions list, then aborting the firmware update.

18. The method of claim 17, further comprising:

receiving, by the processor, a notification of a firmware 15
version vulnerability associated with a first firmware version;

creating, by the processor, a new entry for the first firmware version; and

adding the new entry to the revoked versions list. 20

19. The method of claim 18, wherein the new entry is signed by a private key prior to being added to the revoked versions list.

20. The method of claim 17, wherein the new revoked firmware version in the firmware update is for versions prior 25
to a current firmware image.

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